

Models of Higher Brain Function

Lecture Notes

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Preface

These notes study higher brain function as a set of modeling problems: perception, attention, memory, decision making, and control are treated as latent computations inferred from behavior and neural data. The course therefore moves between three levels of explanation: what problem the organism appears to solve, what algorithm could solve it, and what neural mechanism could plausibly implement that algorithm.

The notes are intentionally mathematical but not notation-driven. Equations are included when they clarify a representational claim, a dynamical mechanism, or an inference procedure. A useful model should compress data, expose its assumptions, and make clear predictions about where it should fail.

Berlin, April 2026

Notation

Dynamical systems

Symbol	Meaning
x_t	State at time t
u_t	Input or control signal
$f(\cdot)$	State transition or update map
τ	Characteristic time constant

Representations and learning

Symbol	Meaning
h	Hidden representation
W	Weight matrix
L	Objective or loss function
∇L	Gradient of the objective

Probability and inference

Symbol	Meaning
$p(z \mid x)$	Posterior distribution
$\mathbb{E}[\cdot]$	Expectation
$\text{Var}[\cdot]$	Variance
$\arg \max$	Optimization operator

1 Course Overview and Modeling Strategy

Higher brain function is a modeling problem before it is a tooling problem. Memory, planning, inference, abstraction, and decision making are all latent processes: we rarely observe them directly, but instead infer them from behavior, neural activity, and task structure. The course therefore asks how to build models whose hidden variables are meaningful, constrained, and experimentally testable.

The working standard is that every model should answer four questions. What is the observable behavior? Which latent variables explain it? What update rule moves those variables through time? What would count as evidence against the model? Keeping these questions explicit prevents the notes from becoming a catalog of architectures without interpretation.

1.1 What makes a model useful?

A model of higher brain function should do more than fit observations. It should compress behavior into a tractable set of assumptions, explain why certain error patterns appear, and suggest new experiments or interventions. In cognitive and systems neuroscience, this usually means balancing three goals:

- computational clarity,
- biological plausibility,
- empirical testability.

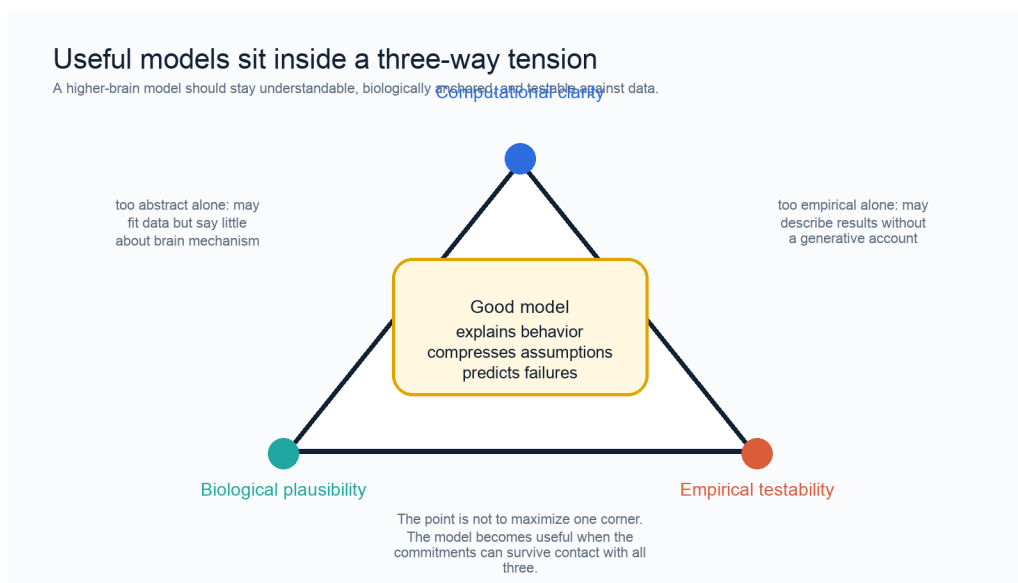


Figure 1: Why useful models live in tension. Strong higher-brain models stay interpretable, stay anchored to plausible mechanisms, and still make contact with measurable data.

Practical heuristic. If a model explains everything equally well, it usually explains nothing. Useful models make sharp commitments about mechanisms, variables, and expected failures.

1.2 A common abstraction stack

Many courses on higher brain function move across a recurring abstraction ladder: behavior and task structure, latent cognitive variables, circuit or algorithmic mechanisms, and normative or optimization principles.

That ladder connects classical cognitive models to modern neural-network and dynamical-systems viewpoints. Reinforcement learning, attractor dynamics, and recurrent computation all make the same point in different mathematical languages: representation, dynamics, and objective functions have to be studied together.

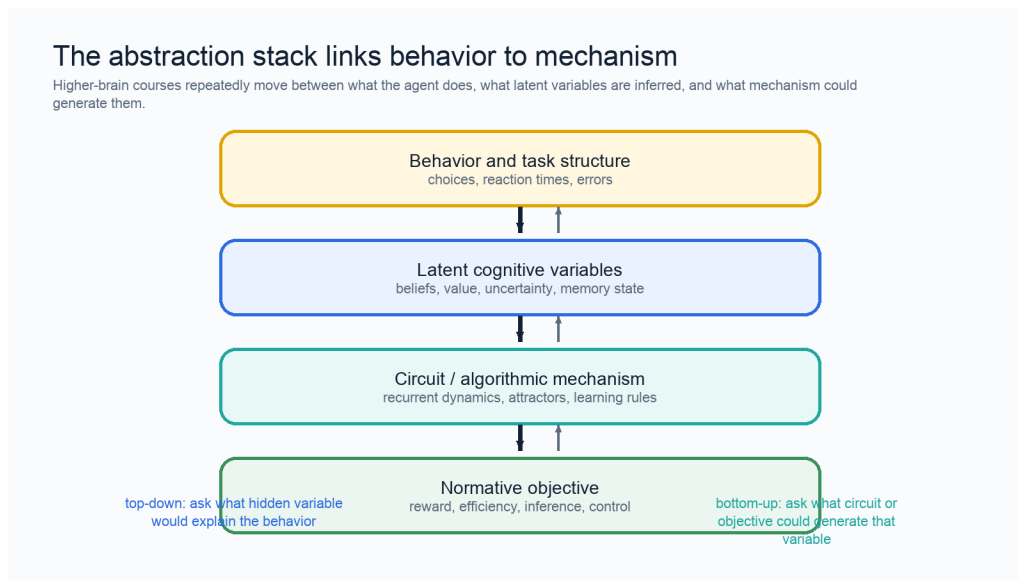


Figure 2: The abstraction ladder used throughout the course. Explanations move downward from observed behavior to hidden variables and mechanism, then upward again to ask whether the mechanism reproduces the behavior.

Generic state-space view.

$$x_{t+1} = f(x_t, u_t; \theta)$$

This form is broad enough to include recurrent neural networks, drift-diffusion style updates, working-memory attractors, and many cognitive-state models.

1.3 Reusable model code

Executable examples should make modeling assumptions visible. Useful code modules include simulation utilities for canonical cognitive tasks, fitting routines for simple latent-state models, plotting helpers for trajectories and phase portraits, and small reading replications that test whether a model reproduces the qualitative phenomenon it claims to explain.

1.4 Next steps

Each conceptual block should end with a compact model checklist: the state variables, the objective or update rule, the measurable predictions, and the most likely confound. This makes later

State-space models make hidden cognition explicit

The latent state summarizes what the system currently knows, while inputs push that state forward in time.

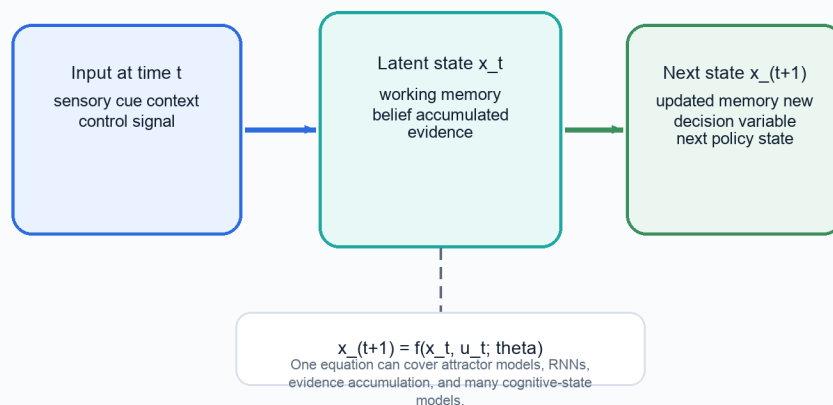


Figure 3: State-space intuition. The hidden state carries forward what the system currently “knows,” inputs perturb that state, and the update rule determines the next internal configuration.

comparisons between attention, memory, decision, and control models precise rather than purely verbal.

2 Lecture 1: Visual Perception and Deep Learning

These notes cover the opening perception lecture of *Models of Higher Brain Function*. The lecture uses vision as the first concrete example of a higher-brain computation: the sensory input is high-dimensional, noisy, and ambiguous, but behavior requires a stable internal description of objects and scenes. Deep neural networks enter the lecture not as a fashionable add-on, but as a way to make the representation problem explicit. They give us a concrete family of models that can solve large-scale object recognition tasks and can then be compared with the ventral visual stream.

2.1 Why vision belongs in a course on higher brain function

The lecture opens with a basic question: what does a visual system need to know about the world? At the level of behavior, the answers are simple enough:

- what is out there,
- where it is,
- and what it is likely to do next.

Those questions already show why vision is not just a matter of registering light intensity. A retinal image is only a pattern of local measurements. Perception requires the system to organize that pattern into useful latent variables such as object identity, position, motion, and behavioral relevance.

2.1.1 Why this problem is harder than it feels

One theme from the lecture is Moravec’s paradox: tasks that feel effortless to humans often turn out to be extremely demanding computationally. Recognizing a cup after it has moved, rotated, changed scale, or appeared under different illumination feels trivial in everyday life, but it is difficult because the raw sensory input changes enormously while the object identity stays the same.

This is why visual perception is treated here as a problem of inference rather than mere detection. The brain must move from a changing image to a useful internal representation of the world.

2.2 Invariant object recognition

The core challenge identified in the lecture is *invariant object recognition*. A useful visual system should keep track of what an object is even when many superficial aspects of the image change.

2.2.1 Why pixels are a bad format for recognition

The lecture follows the argument from DiCarlo and Cox that the raw image is a poor coordinate system for object recognition. Pixel values are highly sensitive to:

- translation across the retina,
- viewing angle,
- distance and scale,
- illumination,
- occlusion,
- and background clutter.

So two pictures of the same object can be very far apart in pixel space, while pictures of different objects may overlap heavily in their low-level image statistics. The slides describe this by saying that object manifolds are “hopelessly tangled up” in the original sensory coordinates.

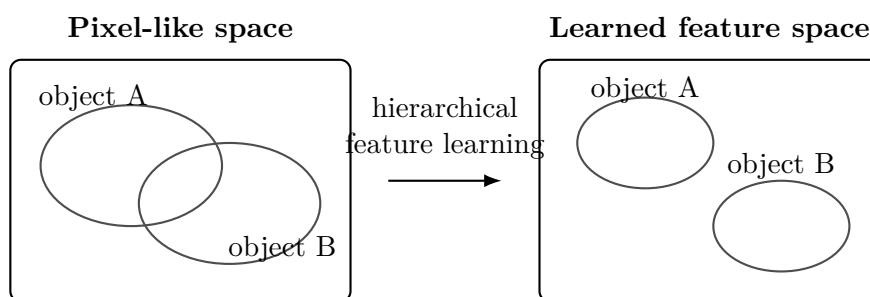


Figure 4: The geometric picture emphasized in the lecture. In raw image coordinates, object manifolds are tangled together. A good representation transforms them into a space where object identity is easier to separate.

2.2.2 What makes a representation useful

The lecture’s answer is that a “good” neural representation should vary gradually with nuisance variables such as position, illumination, or angle, while making object identity easier to read out. In

other words, recognition becomes easier when the representation performs a kind of *disentangling*: factors that should be treated as irrelevant for identification have a smaller effect on the code than they do in the original image.

This idea is important because it shifts the focus away from raw accuracy and toward representational geometry. The real question is not only whether the system labels an image correctly, but whether it transforms sensory input into coordinates that make the task easy for later stages.

2.3 The ImageNet turning point

The lecture then turns from the abstract recognition problem to the modern deep-learning solution. The key historical moment is the ImageNet era. ImageNet supplied a large-scale benchmark with roughly fifteen million labeled images spanning one thousand categories. This mattered because it forced models to solve recognition at a realistic visual scale rather than on small toy datasets.

2.3.1 Why AlexNet changed the conversation

AlexNet’s 2012 ImageNet result is presented as the turning point. Its top-5 error was 16.4%, while the next-best system remained at 26.2%. The significance of that result was not just the leaderboard gap. It showed that a deep, trainable convolutional network could learn useful visual features directly from data instead of depending mainly on hand-engineered descriptors.

The lecture also describes the “magic sauce” behind this success. Several ingredients had to come together at once:

- large labeled datasets,
- data augmentation,
- substantial computing power,
- convolution and weight sharing,
- rectified linear units,
- pooling and downsampling,
- and architectures deep enough to build hierarchical features.

So deep learning did not succeed because of one isolated trick. It succeeded because data, optimization, hardware, and architecture started to reinforce one another.

2.3.2 From AlexNet to ResNet

The lecture places AlexNet in a broader architectural sequence. VGG-style networks increased depth and regularized the use of small convolutions. ResNets then made very deep models trainable by adding skip connections. Instead of asking each block to learn an entirely new transformation, a residual block can learn a correction to the identity map. This helps with vanishing gradients and with the degradation in performance that appears when very deep plain networks become hard to optimize.

The slides make this concrete with the ImageNet benchmark again: ResNets reached depths above one hundred layers and pushed error rates below typical human performance on that dataset. The

lecture's point is not that ImageNet is a perfect model of human vision, but that these models had become competent enough that their internal representations became scientifically interesting.

2.4 What deep networks compute

At the mathematical level, the lecture describes a deep network as a concatenation of linear-nonlinear transformations. A single layer takes an input vector $\vec{x}$, applies an affine transformation, and then a nonlinearity:

$$\vec{y} = \phi(W\vec{x} + \vec{b}).$$

Here W is the weight matrix, $\vec{b}$ is a bias term, and ϕ is a nonlinear activation function.

2.4.1 Why nonlinearity matters

If every layer were purely linear, then stacking many layers would still amount to one overall linear transformation. Depth would not buy us much representational power. Nonlinearity is what lets the network bend and reshape the input space so that later layers can express more abstract and more separable features.

The lecture's qualitative story is straightforward:

- early layers detect simple local patterns such as oriented edges or contrast changes,
- intermediate layers combine those into textures or parts,
- later layers become more useful for category-level distinctions.

This staged abstraction is exactly why deep networks became plausible models of a hierarchical sensory pathway.

2.4.2 Why convolution is useful for images

Convolutional layers add two inductive biases that fit images well.

Local receptive fields. Each filter only sees a small image patch at a time. This reflects the fact that nearby pixels are strongly related and that many informative visual structures are local.

Weight sharing. The same filter is applied at many spatial positions. This makes it possible to detect the same local pattern wherever it appears and reduces the total number of parameters that must be learned.

Pooling and striding then expand the effective receptive field and make the representation less sensitive to small translations. This does not create perfect invariance, but it pushes the representation in the right direction for object recognition.

2.4.3 Backpropagation and the credit-assignment problem

Training is described in the lecture through backpropagation and stochastic gradient descent. A loss function measures the discrepancy between the network output and the target label, and gradients are propagated backward through the chain of nonlinear functions. A typical parameter update has the form

$$w \leftarrow w - \eta \frac{\partial \mathcal{L}}{\partial w},$$

where $\mathcal{L}$ is the loss and η is the learning rate.

This solves the engineering problem of *credit assignment*: it tells the system how each parameter contributed to the final error. The lecture is careful, however, not to overinterpret this as a biological mechanism. Exact backpropagation, large labeled datasets, and perfectly shared convolutional filters are all biologically questionable. So deep networks are useful models of visual computation without being literal copies of cortical learning.

2.5 Deep networks as models of ventral visual cortex

Once deep networks became good at object recognition, the next question was whether their internal representations looked at all like those found in ventral visual cortex. The lecture emphasizes that the comparison should not be made unit by unit. A hidden unit in a trained network is not expected to correspond neatly to one biological neuron, and different random initializations of the same architecture need not preserve any unique one-to-one matching.

2.5.1 Why compare representational geometry

The more useful comparison is at the level of *representational similarity*. For a collection of stimuli, one can compute a response pattern for each stimulus and then ask how dissimilar those patterns are from one another. This yields a representational dissimilarity matrix:

$$\text{RDM}_{ij} = 1 - \text{corr}(r_i, r_j).$$

Here r_i and r_j denote the response vectors evoked by stimuli i and j .

The important point is that two systems can have similar representational geometry even if their individual units are not aligned. This is why representational similarity analysis became so useful in comparing human IT, monkey IT, and deep-network hidden layers.

2.5.2 The RSA construction

Suppose the experiment presents n stimuli. For one system, such as a cortical area or a neural-network layer, collect the responses into a matrix

$$R = \begin{bmatrix} r_1^\top \\ r_2^\top \\ \vdots \\ r_n^\top \end{bmatrix} \in \mathbb{R}^{n \times p}.$$

The row $r_i \in \mathbb{R}^p$ is the population response to stimulus i . In a brain dataset, the p dimensions might be voxels, recording sites, or neurons. In a network, they might be hidden units or feature channels.

RSA then replaces the response matrix by an $n \times n$ matrix of pairwise dissimilarities:

$$D_{ij} = d(r_i, r_j).$$

The function d defines what kind of representational difference is being measured. The most common choice in this lecture is correlation distance,

$$d_{\text{corr}}(r_i, r_j) = 1 - \frac{(r_i - \bar{r}_i \mathbf{1})^\top (r_j - \bar{r}_j \mathbf{1})}{\|r_i - \bar{r}_i \mathbf{1}\|_2 \|r_j - \bar{r}_j \mathbf{1}\|_2}.$$

This quantity ignores the overall response scale and asks whether the two response *patterns* have the same shape across units. A small value means that the two stimuli evoke similar population patterns; a large value means that they are represented far apart.

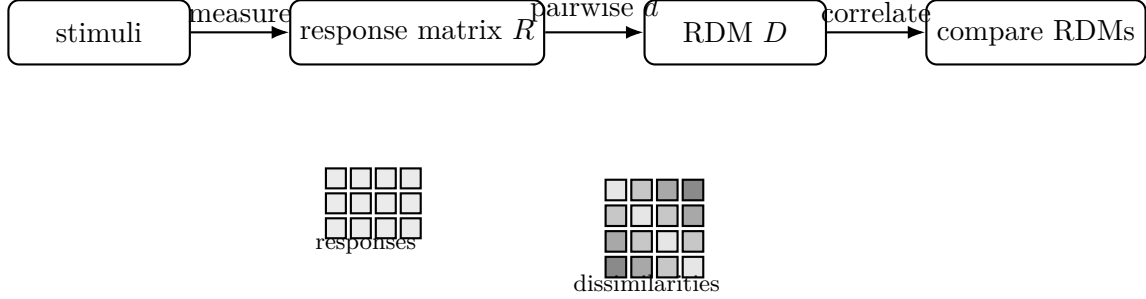


Figure 5: Representational similarity analysis turns high-dimensional response patterns into a matrix of stimulus-by-stimulus distances. The comparison between brain and model happens at the level of these distance matrices.

Other distance choices answer slightly different questions:

- Euclidean distance, $d_{\text{Euc}}(r_i, r_j) = \|r_i - r_j\|_2$, keeps absolute response magnitude and is therefore sensitive to scaling.
- Mahalanobis distance, $d_{\text{Mah}}(r_i, r_j) = (r_i - r_j)^\top \Sigma^{-1} (r_i - r_j)$, downweights noisy or correlated response dimensions through the covariance matrix Σ .
- Rank-based distances ignore exact metric values and ask whether the ordering of near and far stimulus pairs is preserved.

So an RDM is not just a visualization. It is a precise mathematical summary of the geometry induced by a representation on a fixed stimulus set.

2.5.3 Comparing a brain RDM with a model RDM

The second step is to compare two RDMs. Let D^{brain} be the RDM from a cortical area and D^{model} be the RDM from a network layer. Because both matrices are symmetric and have zeros on the diagonal, one usually vectorizes only the upper triangle:

$$v(D) = (D_{12}, D_{13}, \dots, D_{1n}, D_{23}, \dots, D_{n-1,n}).$$

The RSA score can then be written as

$$\rho_{\text{RSA}} = \text{corr}(v(D^{\text{brain}}), v(D^{\text{model}})).$$

In practice this is often a Spearman rank correlation, because researchers may care more about whether the model preserves the ordering of representational distances than about matching every numerical distance exactly.

This is the main conceptual advantage. The number of units in the two systems can be different, and their axes need not correspond. A cortical area might be measured with $p = 500$ voxels and a network layer might have $q = 4096$ features. RSA can still compare them because both are reduced to the same $n \times n$ stimulus geometry.

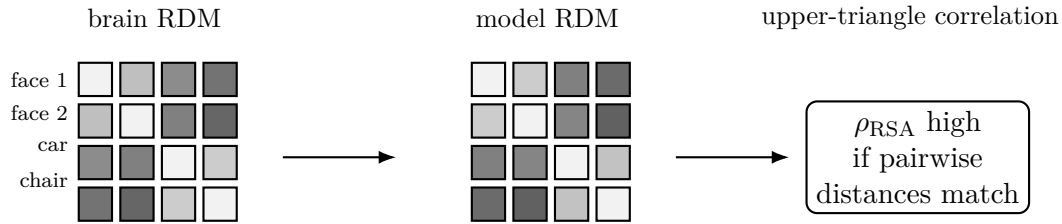


Figure 6: RSA is a second-order comparison. It does not ask whether one model unit equals one neuron. It asks whether pairs of stimuli that are close or far apart in cortex are also close or far apart in the model.

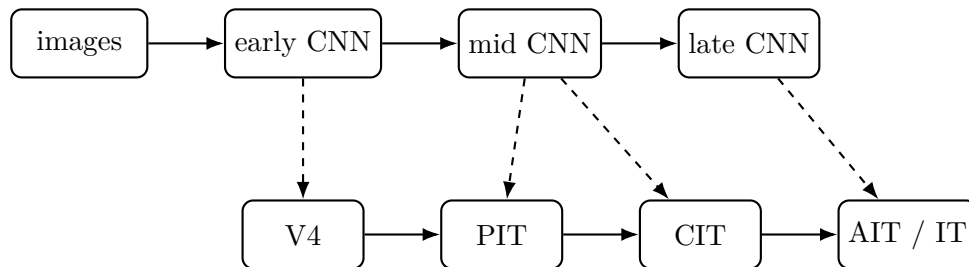


Figure 7: The comparison strategy highlighted in the lecture. Rather than matching units one by one, researchers compare which network layers best predict or resemble activity in successive ventral-stream areas.

2.5.4 How to interpret an RSA result

An RSA match should be read as evidence about representational geometry, not as a claim that the mechanisms are identical. A high RSA score says that the two systems place the tested stimuli in a similar pattern of relative distances. It does not by itself say that the same features were learned, that the same learning rule was used, or that the same computations occur internally.

There are two practical cautions:

- The stimulus set matters. An RDM built from faces, animals, tools, and places tests a different geometry than an RDM built only from small pose changes of one object.
- The noise ceiling matters. If repeated brain measurements are themselves only moderately correlated, then no model can be expected to reach a perfect RSA score.

This is why the lecture treats RSA as a powerful but limited bridge between deep learning and neuroscience. It gives a mathematically clean way to ask whether a model layer has the same *shape of representation* as a cortical area, while still leaving open the mechanistic question of how that representation is produced.

2.5.5 What the comparisons show

The lecture highlights a consistent empirical pattern: supervised deep models tend to explain the geometry of IT responses better than unsupervised alternatives, and models that are better optimized for object recognition often provide better predictions of neural data. This does not mean that cortex literally implements AlexNet or ResNet. It means that solving the right task appears to push artificial systems toward internal representations that partially resemble those found in the

brain.

That is a strong scientific claim. It suggests that task demands and representational constraints are deeply connected: if a system must solve invariant object recognition well, some aspects of its internal geometry may be forced into a similar shape, whether the system is biological or artificial.

2.6 Deep networks as experimental tools

The lecture goes further than model comparison and treats deep networks as practical tools for neuroscience experiments. The Bashivan-style examples make this especially clear. The logic is:

1. train a convolutional network on a large visual task,
2. learn a mapping from model features to recorded neural responses,
3. optimize the input image to maximize the predicted response of a chosen neuron or population,
4. and then test the synthesized image experimentally.

This matters because a differentiable model can be used not only to *explain* old data but also to *design* new stimuli. If the model captures something important about a neuron's preferred feature geometry, then the optimized image should strongly drive that neuron. The lecture discusses V4 experiments in which CNN-optimized images indeed strongly activated recorded neural sites.

So the deep model becomes more than a classifier. It becomes a hypothesis generator for what features a neural population is actually sensitive to.

2.7 What deep networks still miss

The lecture closes with several cautions. Modern deep networks are highly effective at image classification, but that does not make them complete theories of biological vision.

2.7.1 Adversarial fragility

Deep networks can be fooled by adversarial perturbations: small image changes that are nearly imperceptible to humans but cause a dramatic change in the predicted label. This shows that strong benchmark performance does not automatically imply human-like robustness or understanding.

2.7.2 Biological caveats

Several ingredients of standard deep learning remain biologically questionable:

- exact backpropagation,
- extensive supervised labels,
- strict convolutional weight sharing,
- and mostly feedforward inference in systems whose biological counterparts are strongly recurrent.

These caveats do not make deep networks unhelpful. They simply define the proper level of interpretation. Deep networks are useful computational models of visual recognition, and they may capture important aspects of cortical representation, but they are still abstractions.

2.8 Summary of concepts introduced in Lecture 1

The lecture introduced several ideas that should be kept distinct:

Concept	Role in the lecture
Invariant object recognition	keep identity stable across natural image changes
Disentangling	transform tangled image manifolds into easier features
ImageNet	large-scale benchmark that made modern comparison possible
Convolution	local receptive fields plus weight sharing for images
Backpropagation	practical credit-assignment method for deep networks
RSA / RDMs	compare vectorized stimulus-distance matrices across brains and models
CNN-optimized stimuli	use models to generate experimental probes
Adversarial examples	reveal limits of current deep-network robustness

2.9 Conceptual takeaways from Lecture 1

The opening vision lecture establishes the framework used later in the course.

- The main difficulty of vision is not detecting light, but constructing a representation that supports stable object recognition.
- Raw images are a bad format for recognition because nuisance variables strongly distort pixel values.
- Deep networks matter because they provide a concrete way to learn hierarchical feature spaces that partially solve this representation problem.
- The success of deep learning depends on a combination of architecture, data, optimization, and compute rather than on one isolated trick.
- Model evaluation can proceed on several levels at once: recognition performance, similarity to cortical representations, and usefulness for generating new experiments.
- High benchmark accuracy is not the same as biological plausibility or human-like robustness.

These points motivate the broader theme of the course: higher brain function can often be studied by asking what representation a system needs in order to solve a demanding cognitive task.

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3 Lecture 2: Models of Attention

Attention is the first lecture topic in which the course moves from representation to selection. Vision already gave us the problem of constructing useful internal variables from high-dimensional input. Attention asks a different question: when many representations are simultaneously possible, which ones should receive processing priority, stronger gain, memory access, or control over action?

A useful mental model is that attention is a routing policy for limited computation. The visual system cannot treat every location, feature, and object as equally urgent at every moment. It therefore needs mechanisms that amplify some signals, suppress others, and move processing resources from one candidate to the next.

This is why William James’s classical description still feels right at the phenomenological level: attention is selective possession by the mind. But the lecture immediately sharpens that intuition. Attention is not one single faculty. It appears in several forms, and different experiments probe different mechanisms.

The lecture is organized around three levels of description:

- behavioral signatures, especially pop-out and serial search;
- algorithmic models, especially dynamic routing and saliency maps;
- neural circuit models, especially biased competition.

The common problem across all three levels is the same: selection under limited processing capacity.

3.1 What kinds of attention are being discussed?

The slides distinguish several contrasts that are easy to state but important not to collapse.

3.1.1 Overt and covert attention

In overt attention, the sensor itself is moved toward the target. In vision this usually means an eye movement. In covert attention, the eyes remain fixed while processing priority shifts elsewhere in the visual field.

This distinction matters because it separates selection from movement. A subject can keep fixation at one location and still process another location more strongly. The lecture uses this point to motivate attention as a genuine internal control variable rather than just a by-product of gaze.

3.1.2 Endogenous and exogenous attention

The lecture also separates two antagonistic sources of control:

- *endogenous* or top-down attention, which is voluntary and task-guided;
- *exogenous* or bottom-up attention, which is captured by salient stimuli.

Bottom-up attention answers the question “what stands out in the input right now?” Top-down attention answers the question “what is relevant for the current goal?” Much of the lecture is really about how these two influences are combined or made to compete.

3.1.3 Spatial, feature-based, object-based, local, and global attention

Spatial attention prioritizes a location. Feature-based attention prioritizes a property such as redness or a specific orientation. Object-based attention prioritizes all parts that belong to the same object. Local and global attention change the scale of processing, shifting emphasis either to fine structure or to large-scale form.

These are not different buzzwords for the same process. They correspond to different coordinates in which selection can be expressed. A location-based spotlight is the simplest case, but the lecture keeps returning to the fact that useful attention often has to be defined in terms of features, objects, or task templates rather than raw retinal position.

Practical implication. When you read an experiment on attention, first ask what is being selected: a place, a feature, an object, or a task-defined template. That choice determines what kind of model is even plausible.

3.2 The behavioral anchor: pop-out and serial search

The cleanest behavioral contrast in the lecture is between pop-out and serial visual search. Let n denote the number of items in a display, and let $T(n)$ be the reaction time required to find the target.

In a pop-out display, the target differs from distractors by a simple low-level feature such as color or orientation. The search-time slope is close to zero:

$$T_{\text{pop}}(n) \approx T_0 + \epsilon n, \quad \epsilon \approx 0.$$

The constant T_0 is the baseline sensory and motor time, while ϵ is the additional cost per distractor. In pop-out, ϵ is near zero because the target produces an immediate priority peak.

In serial search, the target is defined by a conjunction or by a relation that does not create a unique low-level feature peak. Search time increases with display size:

$$T_{\text{serial}}(n) \approx T_0 + \alpha n, \quad \alpha > 0.$$

Now α measures the extra inspection cost per item. The larger α , the more the system behaves as if it must evaluate objects one by one.

If search is self-terminating and the target is present, the expected number of inspected items is roughly $n/2$. If the target is absent, all n items must be checked:

$$T_{\text{present}}(n) \approx T_0 + \alpha \frac{n}{2}, \quad T_{\text{absent}}(n) \approx T_0 + \alpha n.$$

The formulas are simple, but they make the logic of the experiment explicit. Flat slopes suggest that the display itself creates a dominant selection signal. Positive slopes suggest that the display does not uniquely determine the target, so attention must be deployed sequentially.

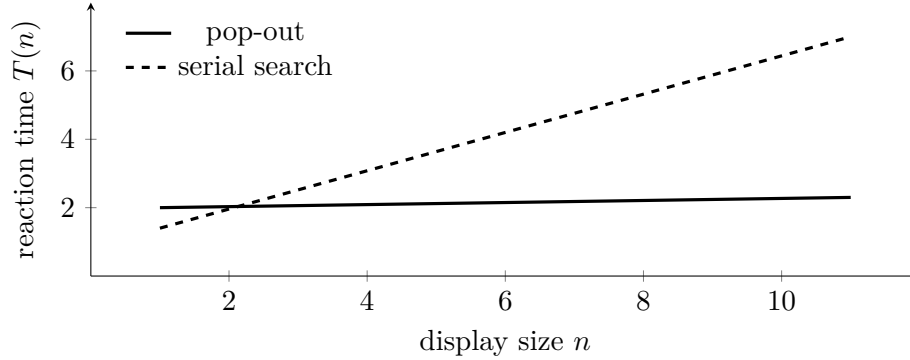


Figure 8: The behavioral signature of attention in search tasks. Pop-out has an almost flat reaction-time slope because one item creates a distinctive priority peak. Serial search has a positive slope because many items must be inspected in sequence.

This distinction is the bridge to all later models. Pop-out means that one candidate already dominates the priority landscape. Serial search means that many candidates remain too similar, so the system needs a policy for shifting selection over time.

3.2.1 Why saliency is about contrast, not raw features

The slides emphasize that saliency is not the same as feature magnitude. A red item is not automatically salient. It is salient when it is red among non-red items. Likewise, an oblique bar pops out when its orientation differs from its neighbors, not simply because it has high orientation energy.

That point is easy to miss, but it is central. Saliency is a contextual quantity. It measures local distinctiveness, not absolute feature value.

Practical implication. To understand whether a display should pop out, compare the target to its surround, not to zero. What matters is feature contrast relative to nearby competitors.

3.3 The spotlight model as a first approximation

The classical spotlight metaphor says that attention selects a restricted spatial region. A compact mathematical version is a spatial gain field $A(x)$ applied to a visual representation $R(x)$:

$$R_{\text{att}}(x) = A(x)R(x).$$

The field $A(x)$ tells us how strongly position x is weighted. If the attended location is x_0 , a narrow focus can be written as

$$A(x) = 1 + g \exp\left(-\frac{\|x - x_0\|^2}{2\sigma_A^2}\right).$$

Here g is the strength of attentional gain, and σ_A is the width of the focus. Increasing g makes the selected region more influential; increasing σ_A broadens the attended area.

This model is useful because it formalizes covert spatial attention: the eyes can remain fixed while a specific retinal region receives enhanced processing. But the lecture also makes clear that the spotlight is incomplete. It only selects in a spatial coordinate system. Feature-based and object-based attention require more flexible selection rules.

3.3.1 Early versus late selection

The spotlight metaphor also frames the old early-versus-late filtering debate. In an early-filter view, unattended information is blocked before deep processing. In a late-filter view, unattended information is processed further but given less weight in report, memory, or action.

The lecture does not resolve that historical debate in an all-or-none way. Instead, the later models suggest a softer conclusion: attention often acts like a graded competition or gain modulation rather than a perfect gate that is either closed or open.

Practical implication. Use the spotlight model as a baseline gain model, not as a full theory. It is best for spatial cueing tasks and weakest when selection follows features, objects, or task templates.

3.4 Beyond space: feature-based and object-based selection

The lecture briefly points to two ways attention escapes a purely spatial description.

In feature-based attention, the system preselects signals that share a relevant low-level feature. If the task is to find a green oblique bar, then green and oblique channels can be upweighted before any final target is chosen. This reduces the candidate set before object-level identification.

In object-based attention, selection spreads within an object. If one part of an object is cued, other parts of the same object may inherit priority. The key variable is now object membership rather than Euclidean position.

A compact way to write the difference is to let the attentional gain depend on a selection variable z :

$$R_{\text{att}}(i) = A(z_i)R(i).$$

If z_i is the location of item i , this reduces to spatial attention. If z_i is a feature label or an object identity, the same gain idea applies in a different reference frame.

This is why the lecture includes both retinal and object-centered coordinates. Useful selection often means transforming information into the coordinate system where the task is simple.

Practical implication. When a task is invariant to retinal position but tied to object identity, expect attention to be expressed in object-centered or feature-centered coordinates rather than as a fixed retinal spotlight.

3.5 Dynamic routing of information

The first explicit model family in the lecture is dynamic routing, following Olshausen and colleagues. Its aim is broader than simple saliency. It tries to explain how attention can dynamically rewire processing so that the same sensory input can be interpreted in different reference frames or routed toward different templates.

A good mental model is a switchboard built from basis connections. The anatomy does not literally rewire on each trial. Instead, a control system changes which pre-existing connection pattern is currently active.

3.5.1 Basis connections and control variables

Let x_j denote input activity and y_i output activity. The effective connection from input unit j to output unit i is written as

$$w_{ij} = \sum_k c_k w_{ij}^{(k)}.$$

The matrices $w_{ij}^{(k)}$ are fixed basis connections, and the control variables c_k choose how much each basis contributes.

This equation says what is being varied. The synaptic basis $w^{(k)}$ is stable; the control coefficients c_k are dynamic. Changing c_k changes the effective routing without changing the underlying hardware.

The output is then

$$y_i = \sum_j w_{ij} x_j = \sum_j \sum_k c_k w_{ij}^{(k)} x_j.$$

So the control variables shape which aspects of the input are transmitted to the output and in what coordinate frame.

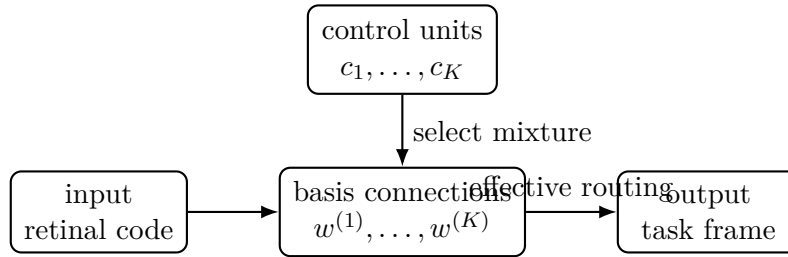


Figure 9: Dynamic routing as a controlled mixture of basis connections. The basis weights are fixed, but the control variables c_k change which routing pattern is active, allowing the same retinal input to be expressed in a task-relevant output frame.

3.5.2 Why the control must be consistent

If all control units were allowed to vary freely, the model could mix incompatible transformations at once. The lecture therefore imposes a consistency constraint: at a given time, the control state should correspond to one coherent routing hypothesis.

In the strongest version, exactly one control unit is active at a time. A simple energy that penalizes simultaneous activation is

$$E_{\text{cons}} = \sum_{i,j} c_i T_{ij} c_j, \quad T_{ij} = \begin{cases} 1, & i \neq j, \\ 0, & i = j. \end{cases}$$

The matrix T encodes mutual incompatibility. If several control units are active together, the energy increases. The minimum is obtained when the control state is sparse and consistent.

Geometrically, this term pushes the control vector toward a corner of the simplex rather than a diffuse average of several contradictory transformations.

3.5.3 Matching the output to a template

The routing is chosen so that the output matches a target pattern P_i . The lecture writes the matching term as

$$E_{\text{match}} = \sum_i y_i P_i.$$

The qualitative role of this term is what matters: it scores how well the routed output aligns with the current template. In practice, one may think of the control system as minimizing a total objective

$$E = E_{\text{match}} + \eta E_{\text{cons}},$$

where η sets how strongly the system insists on a coherent routing state.

The control variables evolve by gradient descent,

$$\tau \dot{c}_k + c_k = -\frac{\partial E}{\partial c_k}.$$

The time constant τ sets how quickly control changes. The gradient term says that control flows downhill in the objective landscape. Different templates produce different downhill directions, so the same input can be routed differently depending on the current goal.

3.5.4 Bottom-up and top-down phases

The lecture describes two phases. In a bottom-up phase, the template is generic, for example a simple brightness blob. The system first finds a plausible candidate based on the sensory input. In a later top-down phase, the template is chosen from memory or task knowledge, and routing shifts toward the most behaviorally likely interpretation.

This is the main conceptual gain of dynamic routing. It offers one framework in which bottom-up and top-down attention are not separate modules but different template choices within the same control dynamics.

Practical implication. For dynamic-routing models, always ask what the control variables are optimizing and in which coordinate frame the template is defined. Those two choices determine whether the model behaves like stimulus-driven capture or goal-driven selection.

3.6 Saliency maps

Saliency-map models answer a narrower question: given an image, which location is most locally distinctive? They do not try to explain all forms of attention. Their main target is early attentional capture and the first few fixation points.

3.6.1 Center-surround feature contrast

Let $I(x, y)$ denote the image. A feature channel k , such as intensity, color opponency, or orientation, is extracted by a multiscale filter:

$$F_k(\cdot, \cdot, s) = h_{k,s} * I,$$

where s is scale, $h_{k,s}$ is a feature-selective filter, and $*$ denotes convolution.

For a center scale c and a larger surround scale s , a feature-contrast map can be written as

$$D_{k,c,s}(x, y) = |F_k(x, y, c) - \text{resize}(F_k(\cdot, \cdot, s))(x, y)|.$$

This is the multiscale version of a center-surround comparison. The center measures what is present here; the surround measures what is present nearby. The absolute difference is large when the location is locally unusual.

A continuous approximation is a difference-of-Gaussians contrast:

$$D_k(x, y) \approx |(G_{\sigma_c} * F_k)(x, y) - (G_{\sigma_s} * F_k)(x, y)|, \quad \sigma_s > \sigma_c.$$

The two scales determine what counts as local context. Increasing σ_s compares the center to a broader neighborhood and changes which structures count as salient.

3.6.2 Normalization favors maps with a clear winner

The Itti-Koch model normalizes feature maps so that maps with one strong peak are emphasized over maps with many equivalent peaks. If M is a contrast map, first scale it to $[0, 1]$:

$$\widetilde{M} = \frac{M - \min M}{\max M - \min M}.$$

Let $m_{\max}$ be the global maximum of $\widetilde{M}$, and let $\bar{m}_{\text{local}}$ be the average value of the other local maxima. A simple version of the normalization is

$$\mathcal{N}(M) = \widetilde{M} (m_{\max} - \bar{m}_{\text{local}})^2.$$

The multiplicative factor is large when one location clearly dominates and small when many locations are equally strong. That is why the same raw feature energy can lead either to pop-out or to ambiguous serial search depending on the competition structure.

3.6.3 From feature maps to a master saliency map

Feature maps are combined into conspicuity maps, and conspicuity maps are combined into a single master saliency map. For channel k ,

$$C_k(x, y) = \sum_{c,s} \mathcal{N}(D_{k,c,s})(x, y).$$

The final saliency map is a weighted sum:

$$S(x, y) = \sum_k w_k \mathcal{N}(C_k)(x, y).$$

Each weight w_k tells us how much a channel contributes. In a purely bottom-up model, these weights are fixed. Limited top-down influence can be introduced by increasing w_k for task-relevant channels, for example if the subject is explicitly searching for a green oblique item.

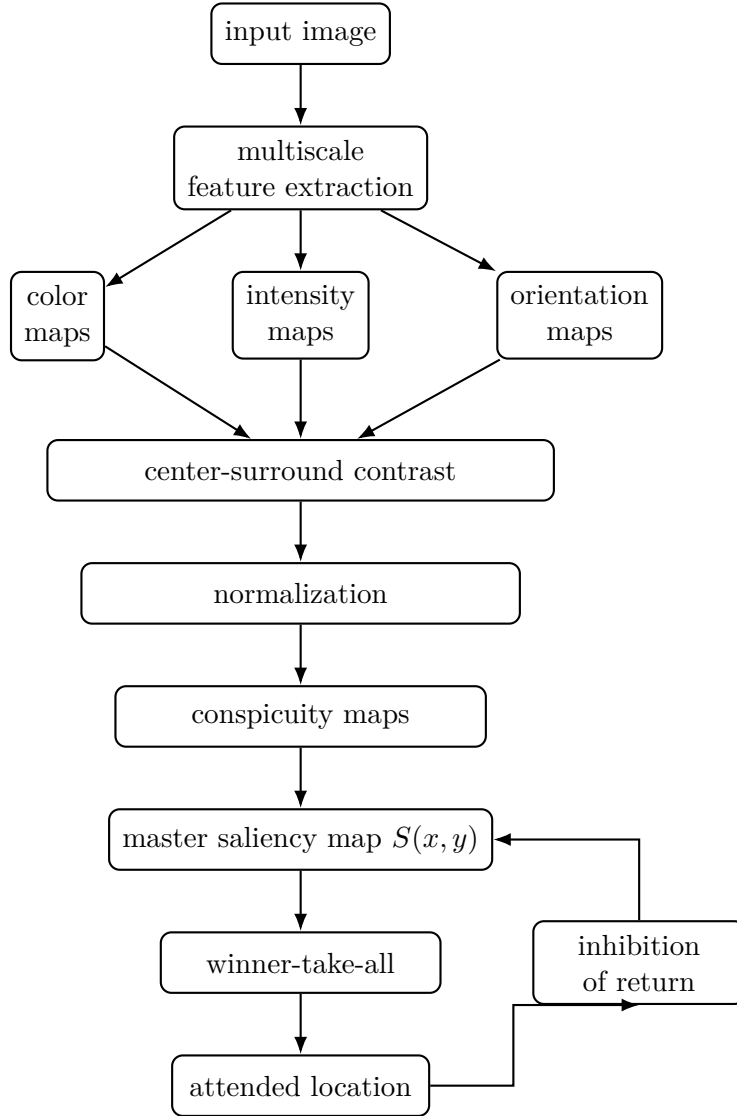


Figure 10: A saliency-map model. The image is decomposed into low-level feature maps, transformed into center-surround contrast maps, normalized, and pooled into a master priority map. Winner-take-all selects the next location, while inhibition of return suppresses recently selected locations.

3.6.4 Winner-take-all and inhibition of return

The current focus of attention is the saliency maximum,

$$\ell_t = \arg \max_{(x,y)} S_t(x, y).$$

The variable ℓ_t is the location selected at time t . To turn a static map into a sequence of attentional shifts, the model adds inhibition of return. After a location has been selected, its neighborhood is transiently suppressed:

$$S_{t+1}(x, y) = S_t(x, y) - \gamma \exp \left(-\frac{\|(x, y) - \ell_t\|^2}{2\sigma_{\text{IOR}}^2} \right), \quad \gamma > 0.$$

The amplitude γ controls how much the previously attended region is suppressed, and σ_{IOR} controls the spatial extent of that suppression. Without inhibition of return, winner-take-all would repeatedly select the same maximum. With it, attention is encouraged to move to the next-best candidate.

3.6.5 What saliency maps do and do not explain

Saliency maps explain attentional capture by low-level features and qualitatively reproduce the pop-out-versus-serial-search distinction. They are especially good at explaining the first few fixation points in simple scenes with low semantic structure.

Their limitation is equally instructive. Later viewing behavior depends strongly on task, memory, reward, and object meaning. A purely bottom-up saliency map cannot explain why a radiologist looks at one patch of an image, why a reader returns to a word, or why personal interest changes gaze.

Practical implication. Use saliency maps when the question is “what captures attention early from the image alone?” Do not expect them to explain long-range, goal-dependent viewing behavior without extra top-down structure.

3.7 Attention on the neural level

The second half of the lecture shifts from where attention goes to how it changes neural responses. The empirical starting point is simple: neural responses are modulated by attention, and the modulation tends to be stronger in higher visual areas such as V4 and IT than in earlier areas such as V1.

A compact rate description is

$$r_i = \phi(g_i s_i + b_i),$$

where s_i is the sensory drive to population i , g_i is multiplicative gain, b_i is an additive bias, and ϕ is a static nonlinearity. This equation is not yet a competition model. It only states the basic ways in which attention can alter a response: by scaling the input, shifting the baseline, or both.

The lecture then asks what happens when several stimuli fall within the same receptive field or otherwise share downstream processing resources.

3.7.1 Stimulus interactions and suppression

If two stimuli are shown together, the response to the pair is generally not the sum of the responses to each stimulus alone. A distractor can suppress the response to the preferred stimulus. The slides summarize this with a suppression index:

$$\text{SI} = 100 \frac{r_{\text{pair}} - r_{\text{best}}}{r_{\text{best}}}.$$

Here r_{best} is the response to the preferred stimulus alone, and r_{pair} is the response when preferred and nonpreferred stimuli are shown together. Negative values mean that adding the distractor reduced the response. The more negative the index, the stronger the competition.

This already suggests that attention cannot be understood as gain on isolated neurons alone. Once several candidate stimuli are active together, the relevant state variable is often the balance between competing populations.

Practical implication. When several stimuli share a receptive field, compare responses to each stimulus alone with the combined response. That residual is the direct signature of neural competition.

3.8 Biased competition

The biased-competition model says that stimuli activate neural populations, those populations inhibit one another, and top-down attention biases the competition toward the task-relevant population. Attention does not create a representation from nothing. It changes which active representation wins.

Let $r_A(t)$ and $r_X(t)$ be firing rates for populations representing stimuli A and X . Let I_A and I_X be bottom-up sensory inputs, and let b_A and b_X be top-down attentional biases. A minimal two-population model is

$$\begin{aligned}\tau \frac{dr_A}{dt} &= -r_A + \phi(I_A + b_A + Er_A - Lr_X), \\ \tau \frac{dr_X}{dt} &= -r_X + \phi(I_X + b_X + Er_X - Lr_A).\end{aligned}$$

The parameter $E > 0$ captures recurrent self-excitation, $L > 0$ captures lateral inhibition, and τ is a time constant. Increasing E stabilizes an active representation; increasing L sharpens competition between the populations.

If attention is directed to A , then $b_A > b_X$. The same physical display can therefore produce different neural responses depending on which object is currently behaviorally relevant.

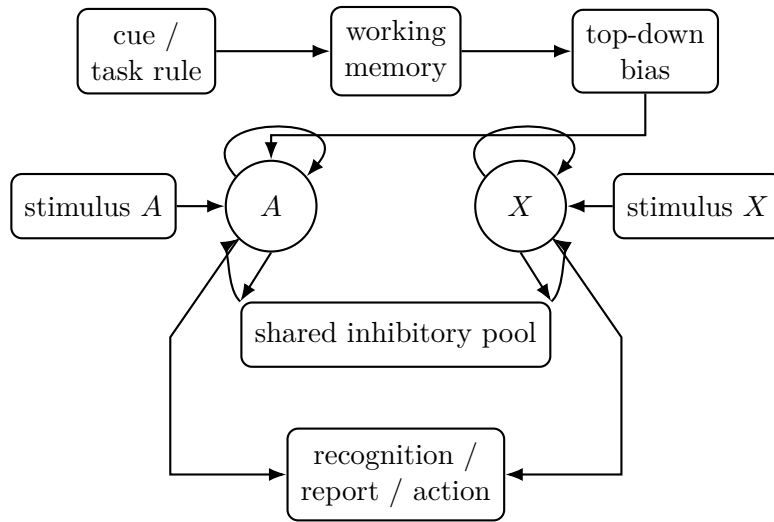


Figure 11: Biased competition. Sensory input activates several populations, recurrent excitation stabilizes each active representation, shared inhibition creates competition, and a top-down signal from the cue or task state biases one competitor.

3.8.1 Why the sum of activity can stay roughly constant

In the linear active regime, where $\phi(z) = z$, the equilibrium obeys

$$(1 - E)r_A + Lr_X = I_A + b_A,$$

$$Lr_A + (1 - E)r_X = I_X + b_X.$$

Adding the equations gives

$$(1 - E + L)(r_A + r_X) = I_A + I_X + b_A + b_X.$$

If total sensory input and total bias are roughly fixed, then the sum of the two firing rates is approximately constrained. This matches the lecture's qualitative point that competition often redistributes activity rather than simply increasing everything at once.

Subtracting the equations gives

$$(1 - E - L)(r_A - r_X) = (I_A - I_X) + (b_A - b_X).$$

This equation is the core of biased competition. The difference in top-down bias directly changes the difference between the competing rates. Attention therefore tilts the balance between active representations.

3.8.2 Cueing, delay activity, and target selection

The Chelazzi-style experiments from the lecture use a cue, a delay, and then a later display containing target and distractor stimuli. The key result is that neural responses to the same physical search array depend on which target was cued beforehand.

The model interpretation is straightforward:

- the cue establishes a working-memory state;
- this state persists during the delay;
- the later display activates multiple sensory populations;
- recurrent excitation and shared inhibition create competition;
- the remembered target supplies a bias that favors one competitor.

This is why the lecture pairs attention with working memory. The bias term is not mysterious. It is often the currently maintained task template.

3.8.3 Divisive normalization as a related formulation

A modern way to write competitive response modulation is divisive normalization:

$$r_i = R_{\max} \frac{(g_i I_i)^n}{\sigma^n + \sum_j (g_j I_j)^n}.$$

The numerator is the drive to population i , amplified by attentional gain g_i . The denominator pools activity across populations, with σ preventing division by zero and setting the operating range. Increasing g_i boosts population i , but because all populations share the denominator, the effect is inherently relative.

This equation is useful because it turns a verbal claim into a quantitative one: attention changes responses under a normalization constraint. It is therefore another way of expressing biased competition in a population code.

Practical implication. When interpreting attentional modulation in neural data, look not only for gain increases but for redistribution under a shared normalization pool. That is often the real computational signature.

3.9 Relation to machine-learning attention

The lecture briefly connects biological attention to modern machine-learning attention. Transformer self-attention computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V.$$

For token i , this becomes

$$y_i = \sum_j \alpha_{ij} v_j, \quad \alpha_{ij} = \frac{\exp(q_i^\top k_j / \sqrt{d})}{\sum_m \exp(q_i^\top k_m / \sqrt{d})}.$$

The weights α_{ij} are normalized routing coefficients. They determine how strongly value vector v_j contributes to the updated representation y_i .

The analogy to biological attention is useful but limited. In both cases, information is selectively weighted rather than uniformly pooled. But transformer attention is not a literal saliency map or a cortical circuit. It has no necessary notion of fixation, inhibition of return, or cue-maintained working memory. The shared idea is normalized context-dependent routing, not identity of mechanism.

Practical implication. The productive comparison is computational rather than anatomical: both systems allocate limited representational influence through normalized weights, but they implement that allocation in very different physical and dynamical ways.

3.10 How the three model families fit together

The lecture is easiest to remember if the three main model families are assigned different explanatory targets.

Model family	Main question it answers
Dynamic routing	How can control variables reconfigure processing so that the same input is interpreted in different task-relevant reference frames?
Saliency maps	Which image location should capture attention first on the basis of low-level feature contrast?
Biased competition	How does attention change neural responses when several stimuli are active and compete within a shared circuit?

Seen this way, the models are complementary rather than mutually exclusive. A task can begin with bottom-up saliency, continue with top-down routing by a template, and end with biased competition inside a cortical population.

3.11 Summary of concepts introduced in Lecture 2

Concept	Role in the lecture
Covert attention	priority shift without an eye movement
Endogenous attention	top-down, voluntary control by the current task
Exogenous attention	bottom-up, stimulus-driven capture
Pop-out	near-flat reaction-time slope caused by a unique low-level priority peak
Serial search	positive reaction-time slope when attention must inspect items sequentially
Spotlight model	spatial gain field over a visual representation
Dynamic routing	controlled reconfiguration of effective connectivity through control variables
Saliency map	master priority map built from normalized feature contrasts
Center-surround contrast	local feature value compared with surrounding context
Inhibition of return	suppression of recently attended locations to support sequential exploration
Suppression index	measure of response reduction when competing stimuli share a receptive field
Biased competition	recurrent competition tilted by top-down task or memory signals
Divisive normalization	shared denominator that turns attentional gain into relative response modulation
Transformer attention	normalized routing analogy, not a literal biological mechanism

3.12 Conceptual takeaways from Lecture 2

- Attention is best treated as a family of selection mechanisms, not as one process.
- Saliency depends on contrast relative to context, not on raw feature magnitude.
- Pop-out and serial search differ because their priority landscapes differ.
- Dynamic routing explains how top-down and bottom-up control can be expressed as different template choices over a shared set of basis connections.
- Saliency-map models explain early attentional capture but are limited when task meaning and memory dominate behavior.
- Biased competition explains why the same stimulus can evoke different neural responses depending on which object is currently relevant.
- In neural populations, attention often redistributes activity under a competition or normalization constraint rather than simply increasing all firing rates.
- The bridge to modern machine learning is normalized routing; the gap is that biological attention is embodied, recurrent, and tied to perception and action.

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4 Lecture 3: Multisensory Integration

This lecture joins two themes that look different on the surface but share the same computational core. The first theme is *perceptual bistability*: one fixed sensory input can support several perceptual interpretations, and the perceived state can switch over time. The second theme is *multisensory integration*: several noisy sensory measurements are combined into one estimate of a hidden property of the world. In both cases, the key variable is not the stimulus itself but the latent hypothesis the brain currently endorses.

A helpful summary is this: bistability emphasizes *competition between interpretations*, whereas multisensory integration emphasizes *reliability-weighted fusion of cues*. The lecture therefore extends the course-wide idea that perception is inference over hidden variables rather than a literal copy of the sensory input.

4.1 Perceptual bistability as competition between interpretations

Bistable perception occurs when a single stimulus is compatible with more than one stable interpretation. Classic examples include ambiguous figures and binocular rivalry. In binocular rivalry, different images shown to the two eyes do not simply fuse into one averaged percept. Instead, one percept dominates for a while, then another takes over.

A minimal population model introduces one activity variable for each candidate percept. Let r_A and r_B denote the activities of populations representing percepts A and B . A standard competitive architecture is

$$\begin{aligned}\tau_r \dot{r}_A &= -r_A + \phi(I_A - \beta r_B - a_A + \xi_A), \\ \tau_r \dot{r}_B &= -r_B + \phi(I_B - \beta r_A - a_B + \xi_B).\end{aligned}$$

Here I_A and I_B are stimulus drives, $\beta > 0$ is mutual inhibition, a_A and a_B are adaptation terms, and ξ_A, ξ_B are noise terms.

Each piece has a clear interpretation:

- the sensory drives favor one interpretation or the other;
- mutual inhibition prevents both interpretations from winning strongly at the same time;
- adaptation slowly weakens the currently dominant percept;
- noise makes the timing of switches variable.

The lecture uses this family of models to explain why switching can happen even when the external stimulus is constant.

4.2 Mechanism 1: adaptation-driven switching

The most direct deterministic explanation for perceptual switching is adaptation. Whichever percept dominates for long enough accumulates an internal penalty. This can be interpreted either as neuronal fatigue or as synaptic depression, but the computational effect is similar: persistent dominance becomes self-limiting.

A simple adaptation dynamics is

$$\tau_a \dot{a}_A = -a_A + \alpha r_A, \quad \tau_a \dot{a}_B = -a_B + \alpha r_B, \quad \tau_a \gg \tau_r.$$

The crucial point is the separation of time scales. The rates r_A, r_B respond quickly, while the adaptation variables a_A, a_B evolve slowly. If percept A dominates, then a_A gradually builds up and eventually reduces the net drive to A enough for percept B to take over.

This already explains an important lesson from the lecture: perceptual switching need not require any change in the stimulus. Hidden internal state variables can force alternation under fixed input.

4.2.1 Oscillator intuition

When adaptation is strong and noise is weak, the system behaves like an oscillator. One percept dominates, adaptation builds against it, the state loses stability, and the competing percept becomes dominant. The same cycle then repeats in the opposite direction.

A reduced intuition can be written using a dominance difference $d = r_A - r_B$ and an adaptation difference $z = a_A - a_B$:

$$\tau_r \dot{d} = F(d, z), \quad \tau_a \dot{z} = -z + \alpha d.$$

Because $\tau_a \gg \tau_r$, the fast perceptual state quickly settles to one branch of the dynamics while the slow adaptation variable drifts until the branch can no longer support dominance. This is why oscillator-style models generate regular alternations.

4.3 Mechanism 2: noise-driven switching

A second explanation says that the system occupies one of several metastable states and that random fluctuations occasionally push it from one to another. In that picture, the noise terms are not minor perturbations; they are essential causes of transitions.

A convenient one-dimensional description introduces an effective energy landscape $E(r)$ over a decision coordinate r :

$$\tau \dot{r} = -\frac{dE}{dr} + \sigma \eta(t),$$

where $\eta(t)$ is white noise. The deterministic term pulls the state toward a local minimum, while the stochastic term can occasionally push the state over the barrier that separates two minima.

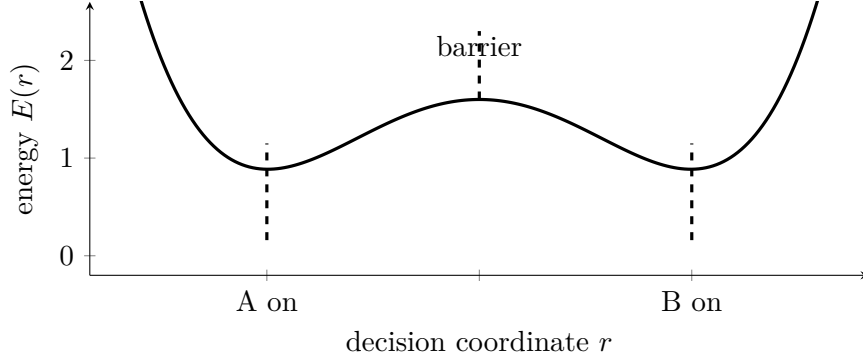


Figure 12: Noise-driven bistability in an energy picture. The two minima correspond to competing percepts. Deterministic dynamics relaxes toward a minimum, while noise occasionally drives the state over the barrier and produces a perceptual switch.

This model explains why strongly fluctuating neural activity can cause switching even when adaptation alone would not force a transition.

4.3.1 Why pure noise predicts exponential durations

If the escape rate from the current attractor is approximately constant in time, then switching is close to a Poisson process. The resulting dominance durations are exponential:

$$P(T > t) = e^{-\lambda t}, \quad p(T) = \lambda e^{-\lambda t}.$$

The parameter λ is a hazard rate: the probability per unit time of leaving the current percept, given that it has persisted so far.

This distribution is memoryless. The chance of switching in the next instant does not depend on how long the current percept has already lasted. The lecture uses this to motivate why pure noise is usually not the full story.

4.4 Why Gamma-like percept durations matter

Empirically, percept durations are often fit better by a Gamma distribution than by a pure exponential law:

$$P(T) = \frac{1}{\Gamma(k)\theta^k} T^{k-1} e^{-T/\theta}, \quad T > 0.$$

Its mean and variance are

$$\mathbb{E}[T] = k\theta, \quad \text{Var}(T) = k\theta^2.$$

For $k = 1$, the distribution reduces to the exponential case. For $k > 1$, the durations become less memoryless and develop a more pronounced mode.

This matters because it tells us something about the hidden mechanism. If durations were nearly periodic, pure adaptation would look sufficient. If they were exactly exponential, pure noise would look sufficient. Gamma-like variability suggests a combination: adaptation changes the underlying hazard over time, while noise still determines the exact switch moment.

4.5 Adaptation plus noise as the robust explanation

The lecture's most convincing synthesis is that adaptation and noise solve different parts of the switching problem. Adaptation introduces a slow drift that makes the currently dominant percept increasingly fragile. Noise introduces variability around that drift.

In the energy picture, one can write

$$E(r, t) = E_0(r) + E_{\text{adapt}}(r, t).$$

As adaptation accumulates, the currently occupied well becomes shallower and the barrier to the alternative percept becomes easier to cross. A useful rate approximation is

$$\lambda(t) \propto \exp\left(-\frac{\Delta E(t)}{\sigma^2}\right),$$

where $\Delta E(t)$ is the current barrier height. As adaptation grows, $\Delta E(t)$ decreases, so the switching hazard increases over time.

The practical implication is that perceptual switching is informative: it reveals hidden state variables such as fatigue, inhibition, and uncertainty that are not visible in the sensory input itself.

4.6 Bayesian sampling views of bistability

A different but compatible perspective says that the percept at any moment is a sample from a posterior distribution over interpretations. Let h denote a latent hypothesis and s the stimulus. Then

$$P(h \mid s) \propto P(s \mid h)P(h).$$

If the posterior over h is multimodal, then several interpretations remain plausible. A sampling process can spend long intervals near one posterior mode and then transition to another.

This provides a conceptual bridge between dynamical-systems language and probabilistic inference. An attractor basin in neural state space can implement one high-probability region of a posterior. In that sense, bistability can be viewed either as competition between dynamical attractors or as sequential exploration of a multimodal posterior.

4.7 Multisensory integration as inference over a hidden cause

The second half of the lecture shifts to cue combination. When vision and touch both provide information about object size, the computational question is not simply how to average the raw measurements. The question is how to infer a hidden world variable from several noisy observations.

Let h denote the hidden property of interest, for example object size. Let s_1 and s_2 denote sensory observations from two modalities. A generative model factorizes the joint distribution as

$$P(s_1, s_2, h) = P(s_1, s_2 \mid h)P(h).$$

Under conditional independence of the modalities given the hidden cause,

$$P(s_1, s_2 | h) = P(s_1 | h)P(s_2 | h).$$

Bayes' rule then gives

$$P(h | s_1, s_2) = \frac{P(s_1 | h)P(s_2 | h)P(h)}{P(s_1, s_2)}.$$

Each term has a direct interpretation:

- $P(h)$ is the prior expectation about plausible hidden states;
- $P(s_i | h)$ describes how modality i behaves if the hidden state were h ;
- the posterior combines prior expectation and current evidence.

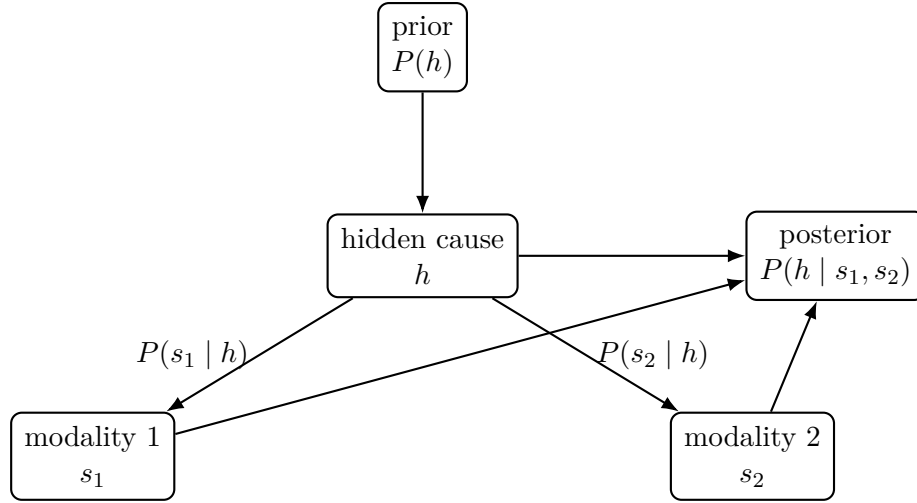


Figure 13: A generative view of multisensory integration. A hidden cause generates noisy observations in several modalities, and inference runs backward from sensory cues to a posterior over the hidden state.

The key lesson is that cue combination is an inference problem, not merely an averaging trick.

4.8 Gaussian cues and the maximum-likelihood estimate

The lecture focuses on the simplest analytically tractable case. Assume a flat prior over the relevant range,

$$P(h) = \text{const},$$

and Gaussian likelihoods,

$$P(s_i | h) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(s_i - h)^2}{2\sigma_i^2}\right).$$

The variance σ_i^2 measures the unreliability of modality i . Small variance means high reliability.

With a flat prior, maximizing the posterior is equivalent to maximizing the likelihood. The log-likelihood is

$$\log P(s_1, s_2 | h) = -\frac{(s_1 - h)^2}{2\sigma_1^2} - \frac{(s_2 - h)^2}{2\sigma_2^2} + \text{const}.$$

Differentiating with respect to h and setting the result to zero gives

$$\frac{h - s_1}{\sigma_1^2} + \frac{h - s_2}{\sigma_2^2} = 0.$$

Solving yields the maximum-likelihood estimate

$$\hat{h} = \frac{p_1 s_1 + p_2 s_2}{p_1 + p_2}, \quad p_i = \frac{1}{\sigma_i^2}.$$

This is one of the main formulas of the lecture. The combined estimate is a weighted average, but the weights are not arbitrary. They are the cue precisions, i.e. inverse variances.

4.8.1 Interpreting the precision weights

Writing the estimate as

$$\hat{h} = w_1 s_1 + w_2 s_2, \quad w_i = \frac{p_i}{p_1 + p_2}, \quad w_1 + w_2 = 1,$$

makes the intuition explicit. If $\sigma_1^2 \ll \sigma_2^2$, then $p_1 \gg p_2$, so modality 1 dominates the estimate. If the two cue variances are equal, then both modalities receive equal weight.

This explains why vision can dominate touch when visual noise is low, and why haptic information can take over when visual noise is increased enough.

4.8.2 Why integration should reduce uncertainty

Combining cues should not only change the mean estimate. It should also make the final estimate more precise. For independent Gaussian cues, precisions add:

$$p_{1+2} = p_1 + p_2, \quad \sigma_{1+2}^2 = \frac{1}{p_1 + p_2}.$$

Because $p_1 + p_2 > p_i$ for either single cue alone, the integrated estimate has smaller variance than either unimodal estimate.

This is the main psychophysical prediction of optimal integration: the bimodal estimate should both reweight the cues correctly and outperform each single cue in discrimination precision.

4.9 Psychometric functions, PSEs, and thresholds

The lecture connects the theory to psychophysics. Suppose a subject judges whether a comparison object is larger than a standard. Across many trials, the proportion of “comparison is larger” responses defines a psychometric curve.

A standard model is

$$P(\text{respond “comparison larger”} \mid x) = \Phi\left(\frac{x - \mu}{\sigma_{\text{psy}}}\right),$$

where x is the comparison value, Φ is the Gaussian cumulative distribution function, μ is the point of subjective equality (PSE), and σ_{psy} controls the slope.

These parameters have direct meanings:

- the **PSE** tells us which comparison value feels equal to the standard, so PSE shifts reveal cue weighting and perceptual bias;
- the **threshold** is related to the inverse slope, so a smaller threshold means finer discrimination.

The theory therefore predicts two measurable effects when cue reliability changes:

- the more reliable cue should receive more weight, shifting the PSE appropriately;
- the bimodal threshold should improve relative to either unimodal threshold.

4.10 The Ernst and Banks result

The lecture discusses the classic visual–haptic experiments of Ernst and Banks. Subjects judged object properties in conditions where visual reliability could be manipulated while haptic reliability remained available. The experiment followed a clean logic:

1. measure unimodal visual thresholds at several visual noise levels;
2. measure unimodal haptic thresholds;
3. convert those thresholds into variance or precision estimates;
4. predict the bimodal weights and thresholds from the maximum-likelihood rule;
5. compare the predictions with observed bimodal behavior.

The central result is that humans often combine visual and haptic information in a statistically near-optimal way. As visual noise increases, the visual weight decreases and the haptic weight increases. The combined threshold is typically close to the value predicted by the sum of unimodal precisions.

This makes the theory experimentally meaningful: the precision-weighted fusion rule is not just mathematically convenient, it is behaviorally testable.

4.11 When the cues disagree too much

The precision-weighted average assumes that the cues arise from the same hidden cause and should therefore be fused. But this assumption can fail. If the visual and haptic cues disagree too strongly, the more important question is not how to weight them but whether they should be integrated at all.

A robust formulation introduces a latent causal variable C indicating whether the cues share a common source:

$$P(h, C \mid s_1, s_2) \propto P(s_1, s_2 \mid h, C)P(h \mid C)P(C).$$

Under $C = 1$, the cues are fused. Under $C = 0$, they are treated as arising from distinct causes and may be kept separate or one may be discounted.

The conceptual lesson is important: optimal fusion is the right rule only *after* deciding that the cues belong together. Causal inference decides whether fusion should happen in the first place.

4.12 A unifying bridge between bistability and cue integration

At first glance, bistability and multisensory integration seem like unrelated topics. One is about switching between percepts and the other about averaging cues. The deeper connection is that both are inference problems over latent variables.

- In bistability, the latent variable is the current perceptual interpretation, and the posterior may be multimodal.
- In multisensory integration, the latent variable is a continuous world property, and the posterior is often treated as unimodal once the cues are assumed to share a common cause.

The posterior shape determines the computational regime. A multimodal posterior encourages switching, sampling, or model selection. A narrow unimodal posterior encourages stable estimation by reliability-weighted averaging. The two halves of the lecture are therefore two versions of the same theme: hidden-state inference under uncertainty.

4.13 Summary of concepts introduced in Lecture 3

Concept	Role in the lecture
Mutual inhibition	prevents incompatible percepts from being strongly active at the same time
Adaptation	slowly weakens the currently dominant percept and promotes switching
Noise-driven escape	random fluctuations push the state across an attractor barrier
Gamma duration law	captures irregular but non-memoryless percept durations
Posterior sampling	interprets percepts as samples from a multimodal posterior
Generative model	specifies how hidden causes produce multisensory observations
Precision	inverse variance; determines cue reliability and weighting
Maximum-likelihood estimate	precision-weighted average of sensory measurements under Gaussian noise
PSE	psychophysical marker of subjective equality and cue weighting
Threshold improvement	behavioral signature that integrated cues reduce uncertainty
Causal inference	decides whether discrepant cues should be fused at all

4.14 Conceptual takeaways from Lecture 3

- Perception is an inference problem over hidden variables, not a direct copy of the stimulus.
- Bistability reveals internal dynamics because several interpretations remain plausible under the same input.
- Adaptation and noise solve different parts of the switching problem and are often both needed.
- Multisensory fusion under Gaussian uncertainty leads to a precision-weighted average rather than an arbitrary blend.
- Optimal integration predicts both the relative *weights* of cues and the reduced *variance* of the combined estimate.
- Large cue discrepancies require causal inference, because the first question becomes whether the cues should be fused at all.

Sources mentioned in Lecture 3

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5 Mathematical Synthesis: Higher Brain Function as Inference

Higher brain function is difficult because the variables of interest are latent. We observe images, actions, response times, choices, spike trains, and population activity; we infer beliefs, memories, task states, goals, and decisions. A good model makes that inference problem explicit.

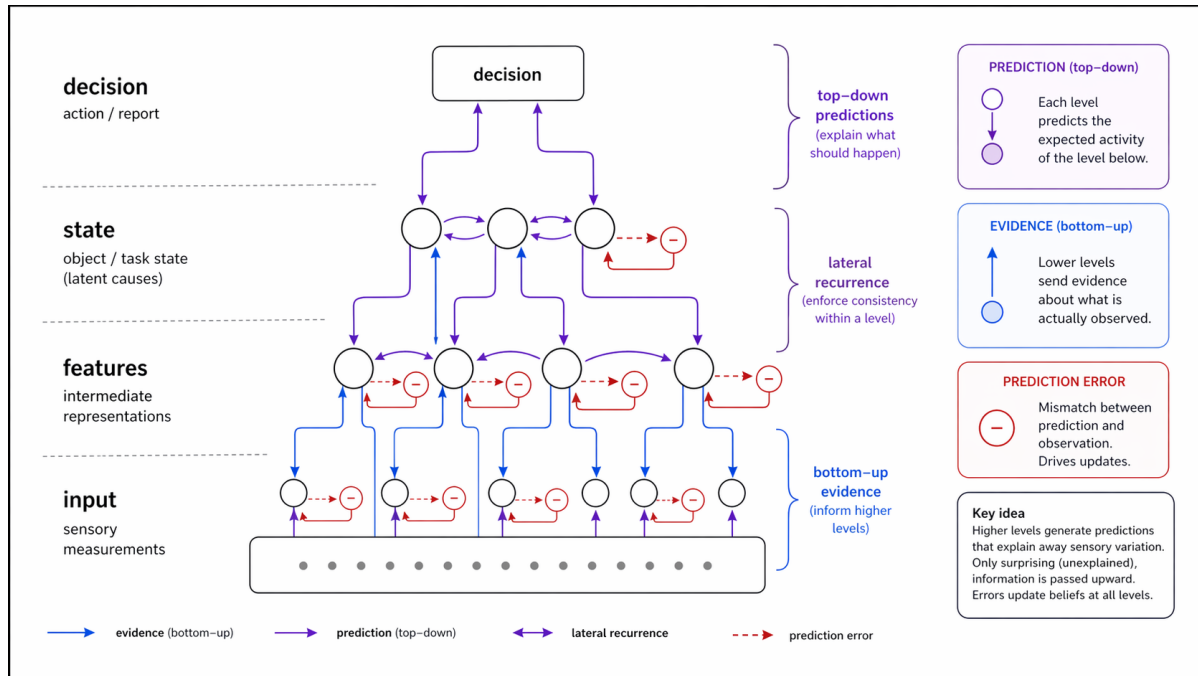


Figure 14: A compact view of hierarchical inference. Bottom-up evidence constrains higher latent states, top-down predictions explain expected sensory structure, lateral recurrence enforces consistency within a level, and prediction errors update the representation.

5.1 Latent States

The basic abstraction is a state-space model:

$$z_{t+1} \sim p(z_{t+1} | z_t, a_t), \quad x_t \sim p(x_t | z_t), \quad a_t \sim \pi(a_t | z_t).$$

Here x_t is observed sensory or behavioral data, z_t is the latent cognitive state, and a_t is an action or report. The model says that cognition is not a direct label attached to the stimulus; it is an inferred state that evolves over time and helps predict both future observations and behavior.

Modeling intuition. The latent state should be just rich enough to make the future conditionally simpler. If knowing z_t makes past details irrelevant for predicting the next observation or action, then z_t has captured the right memory for the task.

5.2 Prediction And Error

In a hierarchical model, higher levels send predictions downward and lower levels send mismatches upward. A minimal predictive-coding objective can be written as

$$E(z) = \frac{1}{2} \|x - g(z)\|_{\Sigma_x^{-1}}^2 + \frac{1}{2} \|z - \mu\|_{\Sigma_z^{-1}}^2.$$

The first term penalizes sensory prediction error. The second term penalizes deviation from prior expectations. Inference changes z to reduce the total error:

$$\dot{z} \propto -\nabla_z E(z).$$

This equation is not just an optimization rule. It gives a mechanistic story for recurrent cortical dynamics: activity settles toward a state that jointly respects sensory evidence and top-down expectations.

5.3 Representations As Coordinates

Perception becomes useful when the representation separates task-relevant variation from nuisance variation. If x is an image and $z = f_\theta(x)$ is a representation, then a downstream decision should be simpler in z -space than in pixel space:

$$y \approx h(z), \quad z = f_\theta(x).$$

Deep networks make this idea concrete. Early layers preserve local structure, intermediate layers combine features, and later layers tend to align more strongly with object or task variables. The scientific question is not only whether the model classifies correctly, but whether its internal geometry matches the invariances and error patterns of biological perception.

5.4 Decisions Couple Belief To Action

Higher-brain models often end with a decision rule. A common normative form is

$$a^* \in \arg \max_a \mathbb{E}[U(a, Z) \mid x_{1:t}].$$

The action is chosen under a posterior belief about the latent state Z . This is the bridge from perception to behavior: the system does not act on the raw stimulus, but on an uncertainty-weighted internal estimate of what the stimulus means for the task.